

Lean Business Model Canvas

GreenGrid AI

An AI-Powered Energy Optimization Platform for Government and Public Sector Infrastructure

Lee Hickey - 21407166

Market Validation

Research Methodology

To validate the GreenGrid AI business concept, primary research was conducted through interviews with a potential user and a domain expert in the public sector energy management and sustainability space. This research aimed to understand current pain points, existing solutions, and receptivity to AI-powered alternatives in the Irish and broader European context.

Interview Participants

Participant A - Participant A oversees energy management for multiple public buildings across Dublin, including civic offices, libraries, and community centres. With many years of experience in facilities management, they are responsible for implementing the Council's Sustainable Energy Action Plan and reporting on building performance to senior management. They expressed significant frustration with current manual processes.

Participant B - Participant B works at an Irish sustainability consultancy firm based in southern region of Ireland that specializes in helping public and private sector organisations achieve their decarbonisation targets. With a background in environmental engineering and many years of experience advising local authorities across Munster, Participant B provides strategic guidance on energy efficiency projects and environmental, social and governance (ESG) compliance. They have worked on over 30 public sector sustainability projects across Ireland.

Key Insights from Interviews

Pain Point Validation: Both participants confirmed that energy waste in public buildings is a significant and ongoing challenge in Ireland. Participant A noted: "We're still relying on quarterly energy bills to understand our consumption. By the time I spot an issue, we've already wasted thousands of euros. The council has ambitious climate targets for 2030, but we don't have the real-time visibility to track whether we're on course." Participant B added on to a similar point from their consultancy perspective: "I

see this across every local authority I work with. They have the political will to decarbonise, but the tools haven't caught up with the ambition."

1. Current Solutions Are Inadequate: Existing alternatives were described as fragmented and insufficient. Participant A noted that "We have a basic building management service (BMS) in our main civic building, but it's essentially a glorified thermostat. No predictive capability, no integration with weather data, nothing automated. For our smaller buildings, we're literally tracking energy use in Excel spreadsheets.

Participant B confirmed this pattern: "The standard approach in Ireland is still annual energy audits. You get a nice PDF report, implement a few recommendations, and then nothing until next year". It's completely reactive, not proactive, and that is my motivation to our slow moving thought process.

2. AI Receptivity and Irish Market Context: There was strong interest in AI-powered solutions, with both participants noting increased openness in the Irish public sector. Participant B observed: "The Climate Action Plan 2024 has put real pressure on local authorities. They're now mandated to reduce emissions by 51% by 2030. That regulatory push is making councils much more receptive to innovative solutions. Three years ago, AI in building management seemed futuristic, now it's necessary."

Participant A agreed with a desire to find an integrative system complex enough to compute their already existing metrics without massive computational cost

3. Integration and Data Security Concerns: Both participants emphasized the importance of working with existing infrastructure. Any solution must work with what their respective workplaces have already got. There are smart meters in most buildings now thanks to SEAI grants, but they're not being used to their full potential due to the lack of integrative network.

Participant B raised the public sector procurement angle: "GDPR compliance and data residency are non-negotiable for Irish local authorities. If you're storing building energy data, it needs to be clear where that data lives and who has access". It is evident that government clients will ask these questions upfront.

4. Willingness to Pay and ROI Expectations: It is no surprise the wealth of this country which is evident through my interactions with both of my participants. Both participants indicated budget availability for such solutions with demonstrable ROI. Participant A confirmed an estimated annual energy budget north of €200,000 so a desirable 10-15% savings, that's €20-30K back into public services. The business case practically writes itself, and it may help meet public sector energy efficiency targets.

It goes without saying that there is more to be done within this area with major companies such as GridBeyond, Positive Carbon and FeedAlpha conducting excellent work within their respective capacities but not quite exceeding their potential.

Conclusion

The market validation research strongly supports the GreenGrid AI value proposition within the Irish and broader European public sector context. The identified problems are real and pressing, driven by regulatory mandates like Ireland's Climate Action Plan and the EU's Energy Efficiency Directive. Current solutions are insufficient, relying on manual processes and periodic audits rather than continuous intelligence. Both the end-user (Participant A) and domain expert (Participant B) perspectives confirm clear demand for an AI-powered platform that delivers measurable energy savings without requiring significant hardware investment, this is what I believe to be biggest limitation faced by current market implementations.

The insights gathered have directly shaped the product features, particularly the emphasis on integration with existing smart meter infrastructure, GDPR-compliant data handling, transparent AI recommendations, and ROI-focused reporting aligned with Irish public sector energy reporting requirements.

How AI/data science creates value

Value Driver	AI/Data Science Contribution	Business Impact
Cost Reduction	Predictive models optimize energy consumption	10-15% reduction in energy bills
Risk Mitigation	Anomaly detection prevents equipment failures	Reduced downtime and emergency repairs
Compliance	Automated carbon tracking and reporting	Meets SI 426 and EU requirements
Decision Support	Data-driven recommendations for investments	Better allocation of public funds
Scalability	Well implemented ML models improve with more data	Network effects benefit all customers
Speed	Real-time processing vs. manual audits	Immediate insights, not quarterly reports

The Data Flywheel Effect:

